

Section 4 – Centroids

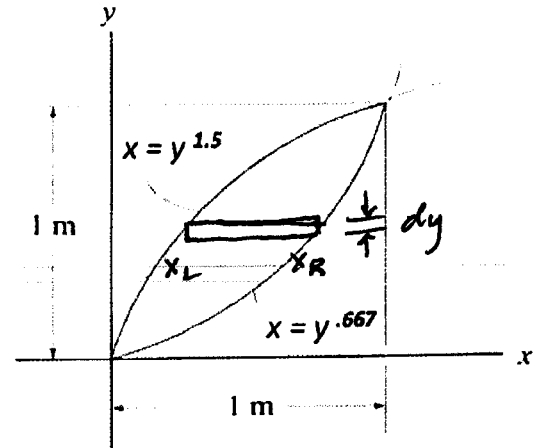
1. (5 pts) When using a **horizontal** differential element of area to solve for the location of $\bar{x}$ for the shaded area, what expression would be correct for dA ?

- A. $(y^{0.667} - y^{1.5}) dy$
- B. $(y^{0.667} + y^{1.5}) dy$
- C. $0.5(y^{1.5} - y^{0.667}) dy$
- D. $(y^{1.5} - y^{0.667}) dy$
- E. $0.5(y^{0.667} + y^{1.5}) dy$

$$dA = (y^{2/3} - y^{1.5}) dy$$

$$\bar{y} = y$$

$$\bar{x} = \frac{x_L + x_R}{2} = \frac{1}{2} [y^{1.5} + y^{2/3}]$$



2. (5 pts) Determine the outside **surface area** of the solid formed by revolving the shaded area 360° about the z axis if $r = 1.5$ m. Do not include the bottom surface/floor of the solid.

- A. 28.3 m^2
- B. 34.6 m^2
- C. 29.9 m^2
- D. 32.0 m^2
- E. 33.0 m^2

$$\theta = 2\pi$$

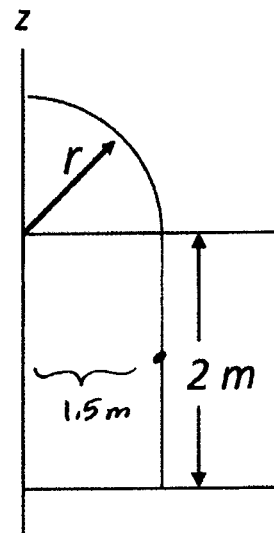
$$\sim L = \frac{2\pi R}{4}$$

$$\frac{2R}{\pi} = 2 \frac{R}{\pi}$$

$$A = 2\pi \left[1.5(2) + \left(\frac{2R}{\pi} \right) \left(\frac{\pi R}{2} \right) \right]$$

$$= 2\pi [3 + 1.5^2] = 2\pi(5.25)$$

$$A = 33.0 \text{ m}^2$$



3. (5 pts) Determine the **volume** of the solid formed by revolving the shaded area 360° about the z axis when $r = 1.5$ m.

- A. 20.0 m^3
- B. 21.2 m^3
- C. 23.2 m^3
- D. 18.1 m^3
- E. 16.7 m^3

$$A = \frac{\pi R^2}{4}$$

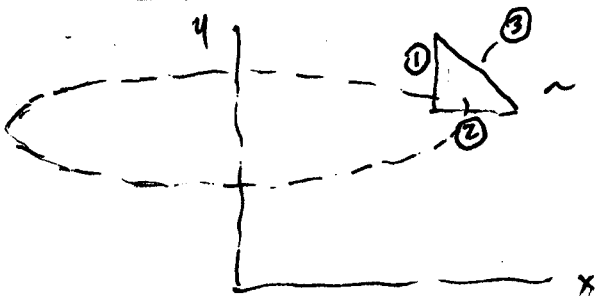
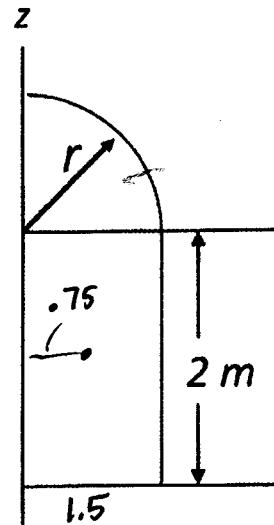
$$\sim \frac{4}{3} \frac{R}{\pi}$$

Don't forget ALL Three lines

for Surface Area

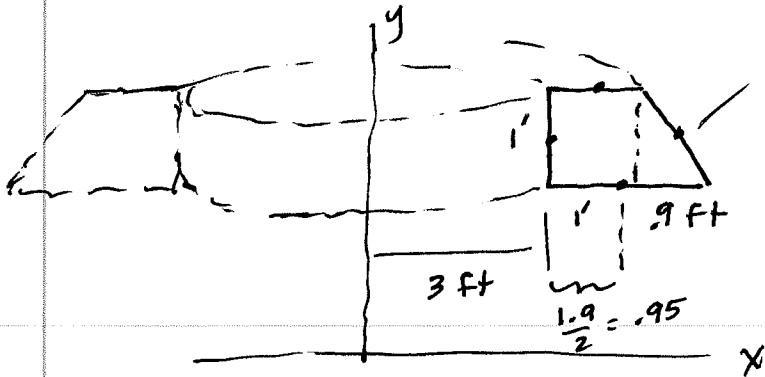
$$V = 2\pi \left[.75(1.5)(2) + \frac{4(1.5)}{3\pi} \frac{\pi(1.5)^2}{4} \right]$$

$$V = 2\pi \left[2.25 + \frac{1.5^3}{3} \right] = 21.2 \text{ m}^3 = V$$



P&G Surface Area + VOLUME

For Surface Area,
don't forget to rotate
all 4 lines



$$L = \sqrt{1 + .9^2} = 1.3454 \text{ ft}$$

4 lines!

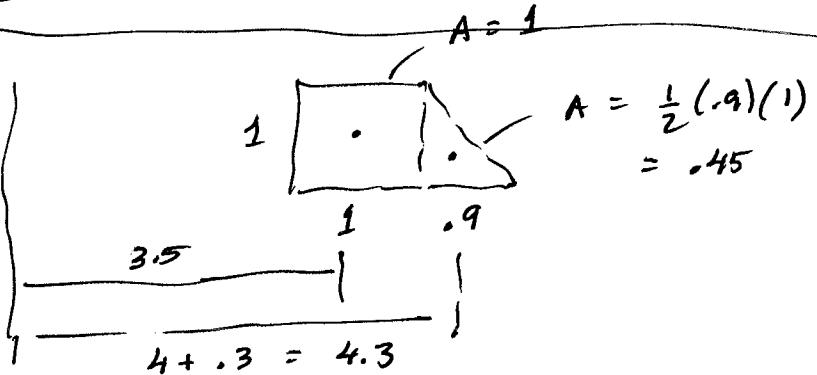
$$L_T = 1 + 1 + 1.9 + 1.3454$$

$$L_T = 5.245 \text{ ft}$$

$$A = 2\pi [3(1) + 3.5(1) + (3.95)(1.9) + (4.45)(1.3454)]$$

$$A = 125.6 \text{ ft}^2$$

VOLUME:



$$A = \frac{1}{2} (.9)(1) = .45$$

$$V = 2\pi [3.5(1) + 4.3(.45)]$$

$$V = 34.15 \text{ ft}^3$$